

TIME-VARYING MEDIATION ANALYSIS FOR INCOMPLETE DATA WITH APPLICATION TO DNA METHYLATION STUDY FOR PTSD

BY KECHENG WEI^{1,a}, FEI XUE^{2,c}, QI XU^{3,d}, YUBAI YUAN^{4,e}, YUEXIA ZHANG^{5,f},
GUOYOU QIN^{1,b}, AGAZ H. WANI^{6,g}, ALLISON E. AIELLO^{7,j}, DEREK E. WILDMAN^{6,h},
MONICA UDDIN^{6,i} AND ANNIE QU^{8,k}

¹*Department of Biostatistics, Fudan University, ^a19211020016@fudan.edu.cn, ^bgyqin@fudan.edu.cn*

²*Department of Statistics, Purdue University, ^cfeixue@purdue.edu*

³*Department of Statistics and Data Science, Carnegie Mellon University, ^dqixu@andrew.cmu.edu*

⁴*Department of Statistics, Penn State University, ^eyvy5509@psu.edu*

⁵*Department of Statistics and Data Science, The University of Texas at San Antonio, ^fyuexia.zhang@utsa.edu*

⁶*College of Public Health, University of South Florida, ^ghussain.kh438@gmail.com, ^hdwildman@usf.edu, ⁱmonica43@usf.edu*

⁷*Department of Epidemiology and Robert N. Butler Columbia Aging Center, Columbia University, ^jaea27@cumc.columbia.edu*

⁸*Department of Statistics and Applied Probability, University of California, Santa Barbara, ^kaqu2@ucsb.edu*

DNA methylation (DNAm) has been shown to mediate causal effects from traumatic experiences to posttraumatic stress disorder (PTSD). However, the scientific question about whether the mediation effect changes over time remains unclear. In this paper we develop time-varying structural equation models to identify cytosine-phosphate-guanine (CpG) sites, where DNAm mediates the effect of trauma exposure on PTSD, and to capture dynamic changes in mediation effects. The proposed methodology is motivated by the Detroit Neighborhood Health Study (DNHS) with high-dimensional and longitudinal DNAm measurements. To handle the nonmonotone missing DNAm in the dataset, we propose a novel longitudinal multiple imputation (LMI) method utilizing dependency among repeated measurements and employ the generalized method of moments to integrate the multiple imputations. Simulations confirm that the proposed method outperforms existing approaches in various longitudinal settings. In DNHS data analysis, our method identifies several CpG sites where DNAm exhibits dynamic mediation effects. Some of the corresponding genes have been shown to be associated with PTSD in the existing literature, and our findings on their time-varying effects could deepen the understanding of the mediation role of DNAm on the causal path from trauma exposure to PTSD risk.

1. Introduction. Posttraumatic stress disorder (PTSD) is a serious mental health disorder that arises after an individual witnesses or experiences traumatic events, such as severe accidents, natural disasters, or warfare. Although the causality of trauma on PTSD has been established from the biological perspective, the trauma-PTSD relation has not been elucidated on the population level of public health data. World Health Organization World Mental Health Surveys show that, although 69.7% of the surveyed population were exposed to at least one traumatic event, the cross-national lifetime prevalence of PTSD was only 5.6% among the trauma-exposed subpopulation (Koenen et al. (2017)). This significant disparity of prevalence between trauma exposure and PTSD implies that there are potential intermediate factors intervening in the trauma-PTSD pathway to explain individual vulnerability to trauma and tendency to develop PTSD. On the other hand, extensive meta-analyses show that the variability in PTSD on the population level is not sufficiently explained by incorporating

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phenotype variables as predictors (Tortella-Feliu et al. (2019)). These limitations in previous PTSD studies lead us to further explore genotype information, such as genetic and relevant molecular variation, as intermediate variables, and identify causal mediation pathways from trauma exposure to PTSD via the transmission of genetic variation.

DNA methylation (DNAm) is a molecular modification that alters gene expression without changing DNA sequence and can be influenced by stress and socioenvironmental factors (Turecki and Meaney (2016), Lussier et al. (2023)). It has been found that aberrant DNAm is associated with a variety of stress-related neuropsychiatric disorders such as depression, schizophrenia, and PTSD (Wani et al. (2021), Wilker et al. (2023)). Thus, DNAm may be a pathway by which traumatic events become biologically embedded and contribute to mental illness. Given the intermediate role of DNAm in mental disorder formulation, studies have suggested that DNAm may also mediate the trauma-PTSD mechanism: trauma increases or reduces DNAm levels, which in turn exacerbate or alleviate PTSD symptoms. For example, Rutten et al. (2018) found that DNAm mediates the relationship between combat trauma and PTSD. Ocean et al. (2022) showed that DNAm in the nuclear factor of activated T cells 1 mediates the exposure to different types of traumatic events and the risk for PTSD. Xue et al. (2022) identified the heterogeneous mediation effects of DNAm in different subpopulations.

From the perspective of mediation methodology, most of the existing mediation analyses on DNAm are cross-sectional based, that is, only estimating mediation pathways at a single time point. However, DNAm levels and PTSD severity typically progress over time, therefore, the relationships among trauma, DNAm, and PTSD may temporally change. For example, Wu et al. (2021) found that different premigration and postmigration stressors significantly affect refugees' mental health at different resettlement stages. Lussier et al. (2023) demonstrated the temporal effects of adversity exposure on genetic variation across childhood and adolescence. Wilker et al. (2023) showed that the association between glucocorticoid receptor gene methylation status and PTSD symptoms is only significant in the early stages of study. Nevertheless, dynamic mediation analysis tools are still not well developed. By performing time-varying mediation analysis on longitudinal data, we can achieve a more comprehensive understanding of how causal pathways between variables in the disease process evolve over time: when the effects occur, diminish, strengthen, weaken, or change direction. Such insights could enrich understanding of the dynamic mediation effects of DNAm on pathways from trauma exposure to PTSD risk and help to develop more targeted and time-sensitive interventions.

There have been some pioneering works about time-varying mediation analysis where exposure, mediator, and outcome are observed over multiple time points. For example, Rijnhart et al. (2022) and Cai et al. (2022) included time-interaction terms in their model to account for temporal effects. Zeng et al. (2021) and Zeng et al. (2023) extended the classical causal mediation method from a functional data analysis perspective. Ge et al. (2023) and Luo et al. (2025) studied dynamic mediation effects under a reinforcement learning framework. Nevertheless, most focus on cases where there is only one or a small number of mediators with completely observed data. Few studies consider scenarios of high-dimensional mediators with incomplete data.

Missing data is ubiquitous in longitudinal study. In the statistical literature, two types of missingness structures are generally considered. One is called "monotone missingness" or "dropout," where a subject may leave the study at a certain time point and never return. On the other hand, "nonmonotone missingness" or "intermittent missingness" refers to when a subject may miss certain visits during follow-up and return at later visits. For our motivating Detroit Neighborhood Health Study (DNHS), which will be introduced later, subjects are supposed to have the methylation status of a large number of cytosine-phosphate-guanine (CpG) sites measured at each follow-up visit. Due to variability of compliance, the DNAm

data exhibits the nonmonotone missingness mechanism, and high-dimensional DNAm values are entirely missing at some visits by the individual. Classical approaches, such as weighting (Chen et al. (2021)), imputation (Jahangiri et al. (2023)), and likelihood-based methods (Tseng et al. (2016)), have been developed to address nonmonotone missingness in longitudinal data. However, these approaches typically consider nonmonotone missingness of the outcome or with a few covariates and may not be directly applicable to our setting where high-dimensional DNAm values are entirely unobserved at some visits.

In this paper we develop a new time-varying mediation method to investigate the temporal mediation effects of DNAm transmission from trauma to PTSD. We propose time-varying structural equation models and adopt a regularization approach to select relevant mediators and identify time-varying effects, which yields several advantages. First, we can identify the important mediators from high-dimensional DNAm values at different time points, in contrast to existing dynamic mediation approaches considering only one or a small number of mediators (Rijnhart et al. (2022)). Second, we can capture the trajectory of mediation effects, even with a small number of time points, while existing approaches require dense time grids (Cai et al. (2022)).

To handle nonmonotone missing DNAm values, we propose a new imputation method called longitudinal multiple imputation (LMI), which has the following major advantages. First, it makes full use of the dependence among repeated measurements, which exploits the inherent correlation structure of longitudinal data, not just based on low-rank structures or trajectory means (Mazumder, Hastie and Tibshirani (2010), Jahangiri et al. (2023)). Second, it can handle high-dimensional mediators being entirely unobserved at some time points, in contrast to existing approaches which only consider missingness of the outcome or with a few covariates (Chen et al. (2021)). Third, it does not require some individuals with complete repeated measurements in the dataset, whereas existing approaches may require that there must be some subjects with complete data or that the data at the first time visit is always observed for all subjects (Jahangiri et al. (2023)).

In simulation studies the proposed method outperforms existing approaches under different missingness structures and different dimensions of mediators, and its performance is robust to different missingness mechanisms. In DNHS data analysis, our method identifies some DNAm CpG sites which exhibit dynamic mediation effects. Specifically, certain CpG sites initially have nonzero mediation effects, but the effects disappear over time. Certain CpG sites do not show mediation effects in early stages, but the effects emerge over time. Some genes (e.g., *PTPRK*), where selected CpG sites (e.g., cg09247979), are located have been reported as differentially expressed genes in previous PTSD meta-analyses (Chitralla, Nagarkatti and Nagarkatti (2016)), and our further findings on their time-varying effects elucidate trauma-PTSD dynamic causal mechanisms.

The rest of the paper is organized as follows. Section 2 describes the background of the DNHS dataset. Section 3 introduces the proposed time-varying mediation models and the nonmonotone missingness structure of mediators. Section 4 presents the proposed approaches for multiple imputation, data integration, and regularization. Section 5 illustrates the detailed implementation and algorithm. Section 6 includes the results of simulation studies. Section 7 applies the proposed method to the DNHS dataset. Section 8 concludes the research with some discussion.

2. Motivating application: Longitudinal trauma, PTSD, and DNAm.

2.1. DNHS data. The Detroit Neighborhood Health Study (DNHS) is a prospective and representative longitudinal cohort study of predominantly African American adults living in Detroit, Michigan (Uddin et al. (2010), Goldmann et al. (2011)). Participants completed a

40-minute structured telephone interview annually between 2008–2012 (i.e., waves 1–5) to assess demographic characteristics, exposure to traumatic events, and mental health status. All participants were given the opportunity to provide a specimen (venipuncture, blood spot, or saliva) for immune and inflammatory marker testing as well as genetic testing of DNA. We analyze the DNAm data with 526 subjects from waves 1, 2, 4, and 5, as there was no full blood draw in wave 3 as in the other waves. We refer to waves 1, 2, 4, and 5 as time points 1–4 throughout the paper.

The exposure variable of interest in our study is the number of traumatic events. In the initial interview (time point 1), participants were asked about their lifetime traumatic experiences using a predefined list of 19 traumatic events (Breslau et al. (1998)). In subsequent interviews (time points 2–4), they were additionally queried about traumatic experiences since their last interview within the same list. At time points 2–4, we incorporate the number of traumatic events experienced since the previous interview to capture the cumulative severity of trauma exposure. The first plot in the left panel of Figure 1 shows the longitudinal trajectories of trauma from 15 subjects.

The outcome variable of interest is PTSD symptom severity, which is measured through the widely used self-report PTSD Checklist (PCL, Ruggiero et al. (2003)). The PCL contains 17 items corresponding to key symptoms of PTSD. In the initial interview (time point 1), participants were asked how much they had been bothered by each symptom, using a five-point scale (1–5), in reference to their worst lifetime traumatic experience, and the summary score for the 17 items was used to characterize the PTSD symptom severity. In subsequent interviews (time points 2–4), they were additionally queried about the worst traumatic experience since their last interview, and the summary score corresponding to it was similarly calculated. At time points 2–4, we choose the largest score calculated in previous and current interviews to capture the cumulative severity of PTSD symptoms. The second plot in the left panel of Figure 1 shows the longitudinal trajectories of PTSD from 15 subjects. We additionally incorporate demographic information, such as age and gender, and blood work, such as CD4 T cells and B cells, measured at baseline.

Existing studies suggest that epigenomic variations can be induced by trauma exposure and accompany the development of stress-related disorders (Rutten et al. (2018), Ocean et al. (2022), Xue et al. (2022)). However, few studies have reported brain-related epigenomic profiles that associate with PTSD risk. In DNHS the methylation levels of 1879 CpG sites from paired blood and brain tissue (Braun et al. (2019)) were assessed. We treat the DNAm levels of these 1879 CpG sites as potential mediators. The methylation level of a CpG site is quantified using the “Beta-value,” which ranges from 0 to 1 and is calculated as the proportion $M/(M + U + 100)$, where M and U represent the signal intensities of methylated and unmethylated probes (Mou, Zhang and Arshad (2022)). The third and fourth plots in the left panel of Figure 1 show the longitudinal trajectories of DNAm levels for two CpG sites from 15 subjects. We observe that the DNAm levels fluctuate within a specific range over the follow-up period, without exhibiting a clear trend or consistent pattern. The average rate of change in DNAm levels for all 1879 CpG sites across the 15 subjects between time points t and t' can be calculated as $CR_{tt'} = \sum_{i=1}^{15} \sum_{j=1}^{1879} (M_{it}^j - M_{it'}^j) / (15M_{it'}^j)$. The corresponding results for CR_{21} , CR_{31} , CR_{41} , CR_{32} , CR_{42} , and CR_{43} are 2.6%, 0%, 1.9%, -0.1% , 1.2%, and 3.2%, respectively.

The traumatic events and PTSD severity are easily assessed so that we have complete repeated measurements for exposure and outcome. However, collecting and testing biospecimens is difficult and costly. Due to variability of compliance, loss to follow-up, and budget restriction, DNAm data could suffer from missingness. In DNHS, DNAm data of some subjects may be available only at certain visits but are unobserved at other visits. The right panel of Figure 1 shows the nonmonotone missingness structure of DNAm. Each blank column

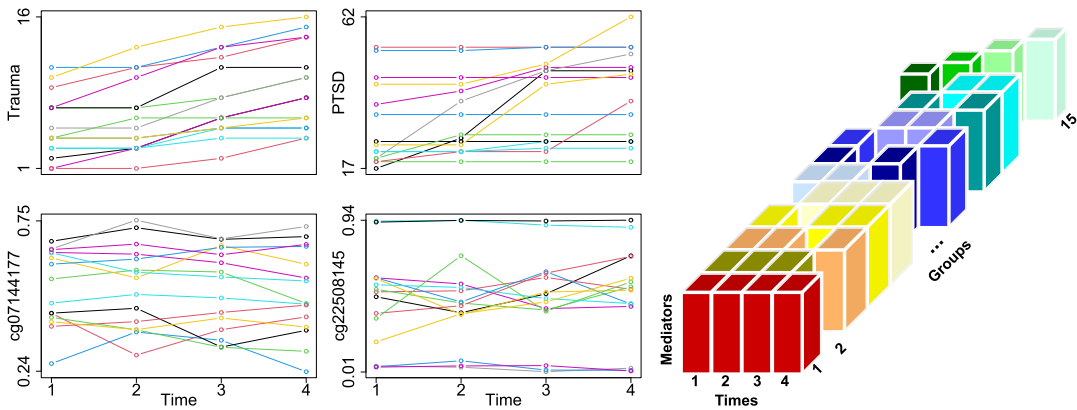


FIG. 1. Left: Longitudinal trajectories of trauma, PTSD, and DNAm of two CpG sites from 15 subjects with complete repeated measurements of DNAm. Right: Nonmonotone missingness structure of DNAm across all time points. Each blank column represents unobserved DNAm, while the colored ones represent observed DNAm.

represents unobserved DNAm, while the colored ones represent observed DNAm. Participants can be divided into 15 groups, where group 1 has four repeated observations at all time points, while group 2 only has observations at time points 1–3, with high-dimensional DNAm data entirely missing at time point 4. The sample sizes of groups 1–15 are (15, 29, 7, 28, 21, 45, 27, 12, 23, 9, 21, 147, 24, 102, 16), respectively. There are 511 out of 526 subjects with missing values and the missingness rate of DNAm is 58.3%.

2.2. *Novelty of our analysis.* Several studies have suggested that DNAm plays a critical role as a mediator in the causal relationship between trauma and PTSD. For instance, Xue et al. (2022) leveraged DNHS data to identify multiple DNAm CpG sites mediating the effect of trauma on PTSD, highlighting that these epigenetic factors can play different mediation roles across different subpopulations. However, their analysis relied on baseline wave to estimate static mediation effects, which is not applicable for discovering potential time-varying genetic contributions to the disease process.

Time-varying effects are a common phenomenon in the genetic regulation of disease progression. Dynamic factors, such as environmental changes, and the natural course of the disease interact intricately with gene regulation mechanisms, suggesting that genetic variants may adapt to changing environmental conditions and show different effects at different stages of the disease (Lussier et al. (2023)). Investigating time-varying genetic effects provides a more nuanced understanding of how genetic factors influence disease processes, helps identify critical time windows for intervention, and facilitates the design of time-sensitive therapeutic strategies.

However, studying time-varying effects poses great challenges. As illustrated in Figure 1, while trauma exposure and PTSD severity tend to increase monotonically over time, the dynamics of DNAm appear irregular, making it difficult to capture their patterns of variation. Moreover, the issues of missingness and high dimensionality within the DNHS dataset further complicate the analysis. These challenges underscore the necessity of developing innovative statistical models and estimation methods to characterize time-varying effects and accommodate complex data structures.

3. **Models.** In this section we propose time-varying structural equation models to study dynamic mediation effects and introduce the nonmonotone missingness structure of mediators.

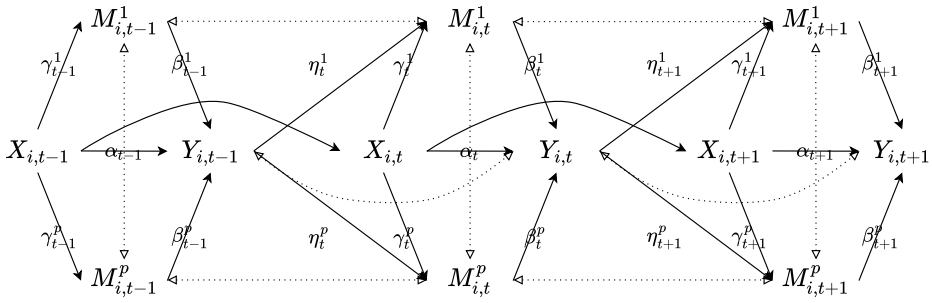


FIG. 2. Dynamic mediation structure among time points $t - 1$, t , and $t + 1$. Solid unidirectional arrows represent causal relationships, while dashed bidirectional arrows indicate correlations.

3.1. Time-varying structural equation models. Let $\mathbf{M}_{it} = (M_{it}^1, \dots, M_{it}^p)^T$ be a $p \times 1$ vector of mediators, where M_{it}^j is the j th ($j = 1, \dots, p$) mediator of the i th ($i = 1, \dots, N$) subject, measured at the t th ($t = 1, \dots, T$) time point. Let $X_{it} \in \mathbb{R}$, $Y_{it} \in \mathbb{R}$, and $\mathbf{Z}_i \in \mathbb{R}^q$ be the longitudinal exposure, the longitudinal outcome, and the demographic variables measured at baseline, respectively. We propose the following structural equation models:

$$(1) \quad Y_{it} = \alpha_t X_{it} + \beta_t^T \mathbf{M}_{it} + \delta^T \mathbf{Z}_i + e_{it},$$

$$(2) \quad \mathbf{M}_{it} = \boldsymbol{\gamma}_t X_{it} + \boldsymbol{\eta}_t Y_{i,t-1} + \boldsymbol{\Gamma} \mathbf{Z}_i + \boldsymbol{\varepsilon}_{it},$$

with $Y_{i0} = 0$, where the coefficients α_t , $\beta_t = (\beta_t^1, \dots, \beta_t^p)^T$, $\boldsymbol{\gamma}_t = (\gamma_t^1, \dots, \gamma_t^p)^T$, and $\boldsymbol{\eta}_t = (\eta_t^1, \dots, \eta_t^p)^T$ measure the time-varying effects at the t th time point, $\boldsymbol{\delta} \in \mathbb{R}^q$ and $\boldsymbol{\Gamma} \in \mathbb{R}^{p \times q}$ measure the time-invariant effects. The random errors e_{it} and $\boldsymbol{\varepsilon}_{it} = (\varepsilon_{it}^1, \dots, \varepsilon_{it}^p)^T$ have zero mean, where e_{it} is independent of X_{it} , $\mathbf{M}_{it}$, and $\mathbf{Z}_i$, and $\boldsymbol{\varepsilon}_{it}$ is independent of X_{it} , $Y_{i,t-1}$, and $\mathbf{Z}_i$. We consider that there are correlations among the residuals e_{it} and $\boldsymbol{\varepsilon}_{it} = (\varepsilon_{it}^1, \dots, \varepsilon_{it}^p)^T$ over $t = 1, \dots, T$ and $j = 1, \dots, p$, which contribute to the correlations among repeated measurements and multiple mediators. We define $\boldsymbol{\Theta} = (\boldsymbol{\theta}^T, \boldsymbol{\vartheta}^{1T}, \dots, \boldsymbol{\vartheta}^{pT})^T$ with $\boldsymbol{\theta} = (\alpha_1, \dots, \alpha_T, \beta_1^T, \dots, \beta_T^T, \delta^T)^T$ and $\boldsymbol{\vartheta}^j = (\gamma_1^j, \dots, \gamma_T^j, \eta_2^j, \dots, \eta_T^j, \boldsymbol{\Gamma}^{jT})^T$, where $\boldsymbol{\Gamma}^j$ is the j th row of $\boldsymbol{\Gamma}$ for $j = 1, \dots, p$.

Figure 2 illustrates the relationships between variables in models (1) and (2) without presenting $\mathbf{Z}_i$, where solid unidirectional arrows represent causal relationships, while dashed bidirectional arrows indicate correlations. For a given time point t , α_t represents the direct effect from the exposure X_{it} to the outcome Y_{it} , and $\gamma_t^j \beta_t^j$ represents the indirect effect through the mediator M_{it}^j . There are correlations among the mediators ($M_{it}^1, \dots, M_{it}^p$). In the context of our real data, the DNAm levels of multiple CpG sites exhibit correlations, reflecting their shared regulatory roles or proximity to similar biological functions (Mou, Zhang and Arshad (2022)).

For adjacent time points $t - 1$ and t , we assume both correlation and causation exist among the exposure, mediators, and outcome. We postulate correlations among repeated measurements of outcome Y_{it} . And similarly, we assume correlations among repeated measurements of mediator M_{it}^j . Regarding the causal relationships across time points, since X_{it} reflects the accumulated level of trauma exposure, it is influenced mainly by the exposure at the previous time point. Based on existing literature, DNAm is primarily influenced by external adverse environmental exposures or disease conditions (Turecki and Meaney (2016)), we specify the causal effect of PTSD condition $Y_{i,t-1}$ on $(M_{it}^1, \dots, M_{it}^p)$ to be $(\eta_t^1, \dots, \eta_t^p)$. Following the Markov assumption (Luo et al. (2025)), we assume that spillover effects among variables occur exclusively at adjacent time points.

Regression coefficients in models (1) and (2) can be linked to causal estimands. Let $Y^{x,m}$ be the potential outcome that would have been observed if X was set to x and $\mathbf{M}$ was set

to $\mathbf{m}$, and $\mathbf{M}^x$ be the potential mediators that would have been observed if X was set to x . Following Bind et al. (2016), we formalize the following assumptions:

- (i) $Y_{it}^{x,m} \perp\!\!\!\perp X_{it} | \mathbf{Z}_i = \mathbf{z}$, (ii) $\mathbf{M}_{it}^x \perp\!\!\!\perp X_{it} | \mathbf{Z}_i = \mathbf{z}$,
- (iii) $Y_{it}^{x,m} \perp\!\!\!\perp \mathbf{M}_{it} | X_{it} = x, \mathbf{Z}_i = \mathbf{z}$, (iv) $Y_{it}^{x,m} \perp\!\!\!\perp \mathbf{M}_{it}^{\tilde{x}} | \mathbf{Z}_i = \mathbf{z}$,

which indicate that there is no: (i) unmeasured exposure-outcome confounding, (ii) exposure-mediator confounding, (iii) mediator-outcome confounding, or (iv) exposure-induced mediator-outcome confounding, where $\tilde{x}$ is the realization of exposure at a different value from x . Corresponding to our DNHS data, to estimate the direct effect of trauma exposure X_{it} on PTSD outcome Y_{it} as well as the indirect effect mediated through DNAm $\mathbf{M}_{it}$, it is essential to control for confounding bias. Assumption (i) requires that all causes of trauma and PTSD are observed and adjusted for. These causes may include demographic information, such as age and gender (Wani et al. (2021)), which have been accounted for in $\mathbf{Z}_i$. Assumption (ii) requires that all causes of trauma and DNAm are observed and adjusted for. These may include demographic information like age and gender (Wani et al. (2021)), which have been accounted for in $\mathbf{Z}_i$. Assumption (iii) requires that all causes of DNAm and PTSD are observed and adjusted for. These may include demographic information, such as age and gender as well as blood work (Occean et al. (2022)), which have been accounted for in $\mathbf{Z}_i$. Assumption (iv) requires that the causes (except for trauma) of DNAm and PTSD are not induced by trauma. This assumption is reasonable, as the demographic information and blood work we control for are intrinsic attributes of individuals. Note that, although $Y_{i,t-1}$ may have a spillover effect on $(M_{it}^1, \dots, M_{it}^p)$, it does not directly affect X_{it} and Y_{it} . Consequently, there is no time-varying confounding in our study, similar to the arguments by VanderWeele and Tchetgen Tchetgen (2017), and thus following Bind et al. (2016), the natural direct effect at time point t for x vs. $\tilde{x}$ is $\mathbb{E}(Y_{it}^{x,m_{it}^x} - Y_{it}^{\tilde{x},m_{it}^{\tilde{x}}}) = \alpha_t(x - \tilde{x})$, and the natural indirect effect at time point t for x vs. $\tilde{x}$ is $\mathbb{E}(Y_{it}^{x,m_{it}^x} - Y_{it}^{x,m_{it}^{\tilde{x}}}) = \boldsymbol{\beta}_t^T \boldsymbol{\gamma}_t(x - \tilde{x})$.

3.2. Nonmonotone missing mediators. In our DNHS data, exposure X_{it} , outcome Y_{it} , and confounders $\mathbf{Z}_i$ are always observed, but the mediators $\mathbf{M}_{it}$ suffer from nonmonotone missingness due to the following reasons: $\mathbf{M}_i = (\mathbf{M}_{i1}, \dots, \mathbf{M}_{iT})^T$ of some subjects may be available only at certain visits but missing at the next time point, and measured again at later visits. Given the nonmonotone missingness, subjects can be divided into R disjoint groups based on different missingness patterns across all time points. Figure 3 illustrates an example where we have $T = 3$ time points and $R = 7$ groups. Each blank column represents unobserved mediators, while the colored ones represent observed mediators. Notice that subjects in group 1 have observed $\mathbf{M}_i$ at all time points, but subjects in group 2 have observed $\mathbf{M}_i$ only at time points 1 and 2.

Define the missing indicator $\mathbf{R}_{it} = (R_{it}^1, \dots, R_{it}^p)^T$, where $R_{it}^j = 1$ for $j = 1, \dots, p$ if $\mathbf{M}_{it}$ is observed at time point t , and $R_{it}^j = 0$ for $j = 1, \dots, p$ if $\mathbf{M}_{it}$ is missing at time point t . In this paper we assume that the missingness mechanism is missing at random (MAR, Little and Rubin (2019)), meaning that the conditional distribution of the missing indicator $\mathbf{R}_i = (\mathbf{R}_{i1}, \dots, \mathbf{R}_{iT})^T$, denoted as $f_{\mathbf{R}_i}(\mathbf{r}_i | X_i, Y_i, \mathbf{Z}_i, \mathbf{M}_i)$, depends only on the exposure $X_i = (X_{i1}, \dots, X_{iT})^T$, the outcome $Y_i = (Y_{i1}, \dots, Y_{iT})^T$, the confounders $\mathbf{Z}_i$, and the observed mediator component $\mathbf{R}_i \mathbf{M}_i$ (Wei et al. (2022)).

Let $\mathcal{H}(r)$ be the index set of subjects in group r and $n_r = |\mathcal{H}(r)|$ be the sample size ($r = 1, \dots, R$). Let $o(r)$ and $u(r)$ be the index sets of time points corresponding to observed and unobserved mediators in group r . Consequently, $\mathbf{M}_{o(r)}$ and $\mathbf{M}_{u(r)}$ represent observed and unobserved mediators in group r . In addition, for each $t \in u(r)$, we define $\mathcal{T}(r, t)$ as the

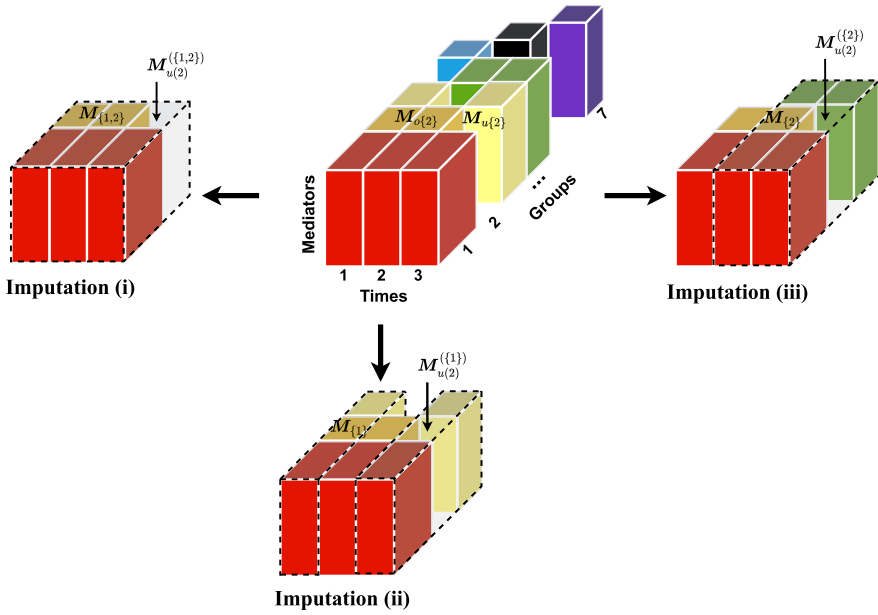


FIG. 3. Middle: Nonmonotone missingness structure of mediators across all time points. Each blank column represents unobserved mediators, while the colored ones represent observed mediators. Surroundings: Multiple imputations for the missing column $M_{u(2)}$ in group 2.

collection of time sets, and each element in $\mathcal{T}(r, t)$ is a subset of $o(r)$. The $\mathcal{T}(r, t)$ can be obtained by the following steps: (i) let $\mathcal{G}(r, t)$ be the index set of the groups where mediators are observed not only at time point t but also at least one time point in $o(r)$; (ii) for each $r' \in \mathcal{G}(r, t)$, let $t_{r'} = o(r) \cap o(r')$; (iii) $\mathcal{T}(r, t) = \bigcup_{r' \in \mathcal{G}(r, t)} t_{r'}$. Note that the definition of $\mathcal{T}(r, t)$ is to prepare for the imputation method later. That is, we can use observed values at each time set $t \in \mathcal{T}(r, t)$ to impute the unobserved values at time point $t \in u(r)$.

4. Methods. In this section we first propose a new imputation method called longitudinal multiple imputation (LMI) to handle nonmonotone missing data. Second, we adopt the generalized method of moments (GMM, Hansen (1982)) to combine the multiple imputations. Third, we propose a regularization approach to identify relevant mediators and estimate time-varying effects. Finally, we employ principal component analysis (PCA) to solve the singularity issue of the weighting matrix in GMM.

4.1. Longitudinal multiple imputation (LMI). In this subsection we propose the LMI approach to handle nonmonotone missing longitudinal data. The main idea is that, due to the correlation between repeated measurements M_t ($t = 1, \dots, T$) from the same subject, we can use the observed values at some time points to predict the unobserved values at other time points. For example, for subjects with observed M_t at time points 1 and 2, we can use some statistical methods, such as regression, to learn the correlation of variables between these two time points. Thus, for some other individuals with observed M_t at time point 1 but missing M_t at time point 2, we can impute the data at time point 2 based on the learned correlation.

Given a group r with a time point $t \in u(r)$, subjects have unobserved M_t^j at this time point but have observed M_t at each time set $t \in \mathcal{T}(r, t)$. We can use observed values at each time set $t \in \mathcal{T}(r, t)$ to impute the unobserved values at time point $t \in u(r)$. Specifically, given a time set $t \in \mathcal{T}(r, t)$, we first estimate $\mathbb{E}(M_t^j | M_t, \mathbf{D})$ using other groups containing observed M_t^j and M_t , where $\mathbf{D}$ contains exposure $\mathbf{X} = (X_1, \dots, X_T)^T$ at all time points,

outcome Y_{t-1} at time point $t - 1$, and confounders $\mathbf{Z}$. Based on the learned correlation, we then impute unobserved M_t^j using $\mathbf{M}_t$ and $\mathbf{D}$. Let $\mathbf{M}_t^{(\ell)} = \{\mathbb{E}(M_t^j | \mathbf{M}_t, \mathbf{D}), j = 1, \dots, p\}$ represent the imputed values based on the time set $\ell \in \mathcal{T}(r, t)$, then we can obtain $|\mathcal{T}(r, t)|$ imputations.

We illustrate the LMI with an example in Figure 3. Specifically, $\mathcal{H}(2)$ refers to the index set of subjects in group 2. Since subjects in group 2 have observed mediators at time points 1 and 2 but have unobserved mediators at time point 3, $o(2) = \{1, 2\}$ and $\mathbf{M}_{o(2)}$ refers to observed mediators, $u(2) = \{3\}$ and $\mathbf{M}_{u(2)}$ refers to unobserved mediators. The collection of time sets $\mathcal{T}(2, 3) = \{\{1, 2\}, \{1\}, \{2\}\}$ since: (i) $\mathcal{G}(2, 3) = \{1, 3, 4\}$; (ii) $\ell_1 = o(2) \cap o(1) = \{1, 2\}$, $\ell_3 = o(2) \cap o(3) = \{1\}$, and $\ell_4 = o(2) \cap o(4) = \{2\}$; (iii) $\mathcal{T}(2, 3) = \bigcup_{r' \in \mathcal{G}(2, 3)} \ell_{r'} = \{\{1, 2\}, \{1\}, \{2\}\}$.

Given group 2 with unobserved $\mathbf{M}_t$ at time point $t \in u(r) = 3$, we can impute the unobserved values in $|\mathcal{T}(r, t)| = |\{\{1, 2\}, \{1\}, \{2\}\}| = 3$ ways: for $j = 1, \dots, p$: (i) for $\ell = \{1, 2\}$, we estimate $\mathbb{E}(M_3^j | \mathbf{M}_{\{1, 2\}}, \mathbf{D})$ using group 1, and then obtain the imputed values $\mathbf{M}_3^{(1, 2)}$; (ii) for $\ell = \{1\}$, we estimate $\mathbb{E}(M_3^j | \mathbf{M}_{\{1\}}, \mathbf{D})$, using groups 1 and 3, and then obtain the imputed values $\mathbf{M}_3^{(1)}$; (iii) for $\ell = \{2\}$, we estimate $\mathbb{E}(M_3^j | \mathbf{M}_{\{2\}}, \mathbf{D})$, using groups 1 and 4, and then obtain the imputed values $\mathbf{M}_3^{(2)}$.

REMARK 1. One novelty of our LMI method is to make full use of the dependence among repeated measurements, which exploits the inherent correlation structure of longitudinal data, not based on stringent low-rank assumptions (Mazumder, Hastie and Tibshirani (2010)) or information-poor trajectory means (Jahangiri et al. (2023)). The LMI method allows high-dimensional variables to be entirely unobserved at some time points, while existing approaches only consider missingness of the outcome or with a few covariates (Chen et al. (2021)). Furthermore, the LMI method allows for the absence of individuals with complete repeated measurements in the dataset, whereas existing approaches may require that there must be some subjects with complete data or that the data at the first time visit is always observed for all subjects (Jahangiri et al. (2023)).

4.2. Combining multiple imputations. In this subsection we propose to combine the information from all LMIs. Since different imputations correspond to different estimating functions, based on moment conditions (Hansen (1982)), we integrate all information through the GMM.

First, we construct the moment conditions corresponding to model (1). For a given group r , $j = 1, \dots, p$, and each time set $\ell \in \mathcal{T}(r, t)$, since unobserved M_t^j at time point $t \in u(r)$ is imputed through $M_t^{j(\ell)} = \mathbb{E}(M_t^j | \mathbf{M}_t, \mathbf{D})$, the covariates $\mathbf{M}_t$ and $\mathbf{D}$ are uncorrelated with residuals of the projection $M_t^j - M_t^{j(\ell)}$. Therefore,

$$\begin{aligned} & \mathbb{E}\{(X_t, \mathbf{M}_t^{(\ell)T})^T (Y_t - \alpha_{0t} X_t - \boldsymbol{\beta}_{0t}^T \mathbf{M}_t^{(\ell)} - \delta_0^T \mathbf{Z})\} \\ &= \mathbb{E}\{(X_t, \mathbf{M}_t^{(\ell)T})^T (e_t + \boldsymbol{\beta}_{0t}^T (\mathbf{M}_t - \mathbf{M}_t^{(\ell)}))\} = \mathbf{0}, \end{aligned}$$

and

$$\mathbb{E}\left\{\sum_{t=1}^T \mathbf{Z}(Y_t - \alpha_{0t} X_t - \boldsymbol{\beta}_{0t}^T \mathbf{M}_t^{(\ell)} - \delta_0^T \mathbf{Z})\right\} = \mathbb{E}\left\{\sum_{t=1}^T \mathbf{Z}(e_t + \boldsymbol{\beta}_{0t}^T (\mathbf{M}_t - \mathbf{M}_t^{(\ell)}))\right\} = \mathbf{0}.$$

The moment conditions are satisfied for the true parameter $\boldsymbol{\theta}_0 = (\alpha_{01}, \dots, \alpha_{0T}, \boldsymbol{\beta}_{01}^T, \dots, \boldsymbol{\beta}_{0T}^T, \delta_0^T)^T$.

We then construct estimating functions based on these moment conditions. For each $i \in \mathcal{H}(r)$, let $\mathbf{M}_{it}^{(\ell)}$ be the imputed values based on time set $\ell \in \mathcal{T}(r, t)$, where $\mathbf{M}_{it}^{(\ell)} = \mathbf{M}_{it}$ for

$t \in o(r)$, and $M_{it}^{j(\ell)} = \mathbb{E}(M_{it}^j | \mathbf{M}_{it}, \mathbf{D}_i)$ for $t \in u(r)$ and $j = 1, \dots, p$. The estimating functions for X_{it} and $\mathbf{M}_{it}^{(\ell)}$ at time point $t \in o(r)$ are $\boldsymbol{\vartheta}_{it}^{(r)}(\boldsymbol{\theta}) = (X_{it}, \mathbf{M}_{it}^{(\ell)T})^T (Y_{it} - \alpha_t X_{it} - \boldsymbol{\beta}_t^T \mathbf{M}_{it}^{(\ell)} - \delta^T \mathbf{Z}_i)$ in which $\mathbf{M}_{it}^{(\ell)} = \mathbf{M}_{it}$, and the estimating functions for X_{it} and $\mathbf{M}_{it}^{(\ell)}$ at time point $t \in u(r)$ are $\boldsymbol{\vartheta}_{it}^{(r)}(\boldsymbol{\theta}) = \{\boldsymbol{\vartheta}_{it}^{(r,\ell)}(\boldsymbol{\theta}), \ell \in \mathcal{T}(r, t)\}$, where

$$(3) \quad \boldsymbol{\vartheta}_{it}^{(r,\ell)}(\boldsymbol{\theta}) = (X_{it}, \mathbf{M}_{it}^{(\ell)T})^T (Y_{it} - \alpha_t X_{it} - \boldsymbol{\beta}_t^T \mathbf{M}_{it}^{(\ell)} - \delta^T \mathbf{Z}_i).$$

The estimating functions for $\mathbf{Z}_i$ are $\mathbf{g}_i^{(r)}(\boldsymbol{\theta}) = \sum_{t=1}^T \mathbf{g}_{it}^{(r)}(\boldsymbol{\theta})$, where $\mathbf{g}_{it}^{(r)}(\boldsymbol{\theta}) = \mathbf{Z}_i (Y_{it} - \alpha_t X_{it} - \boldsymbol{\beta}_t^T \mathbf{M}_{it}^{(\ell)} - \delta^T \mathbf{Z}_i)$ at time point $t \in o(r)$ in which $\mathbf{M}_{it}^{(\ell)} = \mathbf{M}_{it}$, and $\mathbf{g}_{it}^{(r)}(\boldsymbol{\theta}) \in \{\mathbf{g}_{it}^{(r,\ell)}(\boldsymbol{\theta}), \ell \in \mathcal{T}(r, t)\}$ at time point $t \in u(r)$ in which

$$(4) \quad \mathbf{g}_{it}^{(r,\ell)}(\boldsymbol{\theta}) = \mathbf{Z}_i (Y_{it} - \alpha_t X_{it} - \boldsymbol{\beta}_t^T \mathbf{M}_{it}^{(\ell)} - \delta^T \mathbf{Z}_i).$$

To integrate information from all groups, we propose an aggregated vector of estimating functions:

$$(5) \quad \mathbf{g}(\boldsymbol{\theta}) = \{(\mathbf{g}^{(1)}(\boldsymbol{\theta}))^T, \dots, (\mathbf{g}^{(R)}(\boldsymbol{\theta}))^T\}^T,$$

where $\mathbf{g}^{(r)}(\boldsymbol{\theta}) = \sum_{i \in \mathcal{H}(r)} \{(\boldsymbol{\vartheta}_{i1}^{(r)}(\boldsymbol{\theta}))^T, \dots, (\boldsymbol{\vartheta}_{iT}^{(r)}(\boldsymbol{\theta}))^T, (\mathbf{g}_i^{(r)}(\boldsymbol{\theta}))^T\}^T / n_r$. Note that the total number of estimating functions exceeds the dimension of parameters, and the estimating functions from groups with fewer missingness or more precise imputations tend to have smaller variance. To combine all estimating functions in $\mathbf{g}(\boldsymbol{\theta})$ and put more weights on high-quality moments, we propose the following quadratic loss function:

$$(6) \quad G(\boldsymbol{\theta}) = \sum_{r=1}^R (\mathbf{g}^{(r)}(\boldsymbol{\theta}))^T (\mathbf{W}^{(r)}(\boldsymbol{\theta}))^{-1} \mathbf{g}^{(r)}(\boldsymbol{\theta}),$$

with weighting matrix $\mathbf{W}^{(r)}(\boldsymbol{\theta}) = \sum_{i \in \mathcal{H}(r)} \mathbf{g}^{(r)}(\boldsymbol{\theta}) (\mathbf{g}^{(r)}(\boldsymbol{\theta}))^T / n_r$.

For each mediator M^j ($j = 1, \dots, p$) in model (2), we can get the aggregated vector of estimating functions $\mathbf{h}^j(\boldsymbol{\vartheta}^j)$ similar to (5). Detailed formulations are shown in the Supplementary Material (Wei et al. (2025)). We propose the following quadratic loss function:

$$(7) \quad H^j(\boldsymbol{\vartheta}^j) = \sum_{r=1}^R (\mathbf{h}^{j(r)}(\boldsymbol{\vartheta}^j))^T (\mathbf{W}^{j(r)}(\boldsymbol{\vartheta}^j))^{-1} \mathbf{h}^{j(r)}(\boldsymbol{\vartheta}^j),$$

with weighting matrix $\mathbf{W}^{j(r)}(\boldsymbol{\vartheta}^j) = \sum_{i \in \mathcal{H}(r)} \mathbf{h}^{j(r)}(\boldsymbol{\vartheta}^j) (\mathbf{h}^{j(r)}(\boldsymbol{\vartheta}^j))^T / n_r$. The loss function incorporating models (1) and (2) for all subjects is $G(\boldsymbol{\theta}) + \sum_{j=1}^p H^j(\boldsymbol{\vartheta}^j)$.

REMARK 2. For multiple imputed data, different imputations correspond to different estimating functions, all of which satisfy the moment conditions. We use GMM to combine all estimating functions, which adaptively puts more weights to imputations with smaller variance, while still consider the information contained in imputations with larger variance. In contrast, if we only use the imputation with the smallest variance while discarding the others, it may lead to efficiency losses due to the lack of use of available information.

REMARK 3. In constructing the estimating equation for parameter estimation, we ignore the correlation structure among mediators. Specifically, we assume independent correlation matrix in the estimating equation. This is because the correlations among mediators have been implicitly incorporated into our imputation method, where we employ a regression approach to capture the dependency among repeated measurements and multiple mediators jointly, subsequently imputing missing values based on the learned correlations.

4.3. *Regularization.* To select the relevant mediators from high-dimensional variables and to estimate the time-varying effects, we adopt the regularization approach. Specifically, we propose to incorporate three penalty terms to: (i) identify mediators with large mediation effects at each time point, (ii) shrink similar effects at adjacent time points, and (iii) shrink effects of demographic variables. Consequently, the objective function of the proposed method is

(8)
$$L(\Theta) = G(\theta) + \sum_{j=1}^p H^j(\boldsymbol{\vartheta}^j) + P_1(\Theta) + P_2(\Theta) + P_3(\Theta),$$

where

(9)
$$P_1(\Theta) = \sum_{t=1}^T P_{\text{SCAD}, \lambda_1, a}(|\alpha_t|) + \lambda_1 \sum_{t=1}^T \sum_{j=1}^p \left\{ 1 - \frac{1}{(1 + b|\beta_t^j|)(1 + b|\gamma_t^j|)} \right\}$$

is a penalty for direct and indirect effects, with $P'_{\text{SCAD}, \lambda_1, a}(\alpha) = \lambda_1 \{\mathbb{I}(\alpha \leq \lambda_1) + \frac{(a\lambda_1 - \alpha)_+}{(a-1)\lambda_1} \mathbb{I}(\alpha > \lambda_1)\}$ for some $\alpha > 0$, $a > 2$, and $b > 0$,

(10)
$$P_2(\Theta) = \lambda_2 \sum_{t=2}^T |\alpha_t - \alpha_{t-1}| + \lambda_2 \sum_{t=2}^T \sum_{j=1}^p (|\beta_t^j - \beta_{t-1}^j| + |\gamma_t^j - \gamma_{t-1}^j|)$$

is a fusion penalty for time-varying effects, and

(11)
$$P_3(\Theta) = \lambda_3 \left(\|\delta\|_1 + \sum_{j=1}^p \|\mathbf{r}^j\|_1 \right)$$

is a penalty for demographic effects. The regularization parameters λ_1 , λ_2 , and λ_3 control the amount of shrinkage.

The second term in $P_1(\Theta)$ is a bi-level mediation penalty for indirect effects, which links models (1) and (2) by penalizing β_t and γ_t jointly rather than separately, and achieves bi-level variable selection (Huang et al. (2009)). This estimator not only obtains the sparse solution for β_t^j and γ_t^j when $\beta_t^j = 0$ and $\gamma_t^j = 0$, but also obtains the sparse solution for β_t^j or γ_t^j when $\beta_t^j = 0$ or $\gamma_t^j = 0$. On the other hand, it inherits the advantage of the smoothly clipped absolute deviation (SCAD, Fan and Li (2001)) penalty where the function gradually levels off as β_t^j and γ_t^j increase, which avoids overshrinkage. In general, it is possible that not every mediator has different effects at different time points. We use the fused lasso penalty (Tibshirani et al. (2005)) in $P_2(\Theta)$ to shrink similar effects at adjacent time points, which links β_t and γ_t across all time points. The $P_3(\Theta)$ is a lasso penalty (Friedman, Hastie and Tibshirani (2010)) for selecting demographic effects to avoid over-fitting.

4.4. *Principal components of weighting matrix.* The GMM estimators in (6) and (7) may have poor finite sample performance in highly overidentified models, due to imprecise estimation of the weighting matrix $\mathbf{W}^{(r)}(\theta)$ or $\mathbf{W}^{j(r)}(\boldsymbol{\vartheta}^j)$ or under the singularity of the weighting matrix (Doran and Schmidt (2006)). This could be attributed to the large number of estimating functions compared to the relatively small sample size or due to the overlap of information between LMIs. For example, as illustrated in Figure 3, the observed values at time point 1 in group 2 are used for the estimation of both $\mathbf{M}_{u(2)}^{(\{1,2\})}$ and $\mathbf{M}_{u(2)}^{(\{1\})}$, and the observed values at time point 2 in group 2 are used for the estimation of both $\mathbf{M}_{u(2)}^{(\{1,2\})}$ and $\mathbf{M}_{u(2)}^{(\{2\})}$.

To address the singularity issue, we reduce the dimension of $\mathbf{g}^{(r)}$ and $\mathbf{h}^{j(r)}$ ($r = 1, \dots, R$ and $j = 1, \dots, p$) by combining informative estimating functions, for example, using the first several largest principal components. If $\mathbf{W}^{(r)} = \sum_{i \in \mathcal{R}(r)} \mathbf{g}_i^{(r)} \mathbf{g}_i^{(r)T} / n_r$ is singular or close to

singular, we extract the first $u^{(r)}$ principal components $\tilde{\mathbf{g}}^{(r)} = \mathbf{U}^{(r)} \mathbf{g}^{(r)}$ from $\mathbf{g}^{(r)}$, where $\mathbf{U}^{(r)}$ contains $u^{(r)}$ eigenvectors of $\mathbf{W}^{(r)}$ corresponding to the largest $u^{(r)}$ nonzero eigenvalues. Similarly, if $\mathbf{W}^{j(r)} = \sum_{i \in \mathcal{H}(r)} \mathbf{h}_i^{j(r)} \mathbf{h}_i^{j(r)T} / n_r$ is singular or close to singular, we extract the first $u^{j(r)}$ principal components $\tilde{\mathbf{h}}^{j(r)} = \mathbf{U}^{j(r)} \mathbf{h}^{j(r)}$ from $\mathbf{h}^{j(r)}$, where $\mathbf{U}^{j(r)}$ contains $u^{j(r)}$ eigenvectors of $\mathbf{W}^{j(r)}$ corresponding to the largest $u^{j(r)}$ nonzero eigenvalues. The numbers of principal components $u^{(r)}$ and $u^{j(r)}$ ($r = 1, \dots, R$ and $j = 1, \dots, p$) can be selected based on the information criterion proposed by [Cho and Qu \(2015\)](#) to capture sufficient information from the estimating functions. Note that if $\mathbf{W}^{(r)}$ or $\mathbf{W}^{j(r)}$ is not singular, $\mathbf{U}^{(r)}$ or $\mathbf{U}^{j(r)}$ is an identity matrix. Consequently, the final objective function is

$$(12) \quad \tilde{L}(\Theta) = \tilde{G}(\theta) + \sum_{j=1}^p \tilde{H}^j(\theta^j) + P_1(\Theta) + P_2(\Theta) + P_3(\Theta),$$

where the quadratic form $\tilde{G}(\theta) = \sum_{r=1}^R (\tilde{\mathbf{g}}^{(r)}(\theta))^T (\tilde{\mathbf{W}}^{(r)}(\theta))^{-1} \tilde{\mathbf{g}}^{(r)}(\theta)$ with transformed weighting matrix $\tilde{\mathbf{W}}^{(r)}(\theta) = \sum_{i \in \mathcal{H}(r)} \tilde{\mathbf{g}}_i^{(r)}(\theta) (\tilde{\mathbf{g}}_i^{(r)}(\theta))^T / n_r$, and the quadratic form $\tilde{H}^j(\theta^j) = \sum_{r=1}^R (\tilde{\mathbf{h}}^{j(r)}(\theta^j))^T (\tilde{\mathbf{W}}^{j(r)}(\theta^j))^{-1} \tilde{\mathbf{h}}^{j(r)}(\theta^j)$ with transformed weighting matrix $\tilde{\mathbf{W}}^{j(r)}(\theta^j) = \sum_{i \in \mathcal{H}(r)} \tilde{\mathbf{h}}_i^{j(r)}(\theta^j) (\tilde{\mathbf{h}}_i^{j(r)}(\theta^j))^T / n_r$.

5. Computation. In this section we provide the computational details of the proposed method. We first describe the implementation of LMI from Section 4.1. Then we introduce an information criterion to select the numbers of principal components from Section 4.4. Next, we present a smoothing technique to handle the nonseparable fusion penalty from Section 4.3. Afterward, we introduce the overall optimization algorithm of the proposed method for a given set of regularization parameters $(\lambda_1, \lambda_2, \lambda_3)^T$ from Section 4.3. Finally, we propose an information criterion to select these regularization parameters.

When conducting LMI, the conditional expectation $\mathbb{E}(M_t^j | \mathbf{M}_t, \mathbf{D})$ is required to be estimated for each time set $t \in \mathcal{T}(r, t)$, time point $t \in u(r)$, group $r = 1, \dots, R$, and dimension $j = 1, \dots, p$. In this paper we adopt the lasso-regularized linear model ([Friedman, Hastie and Tibshirani \(2010\)](#)) to account for high dimensionality, where the regularization parameter is selected by cross-validation.

To capture most of the information from the estimating functions $\mathbf{g}^{(r)}$ and $\mathbf{h}^{j(r)}$ ($r = 1, \dots, R$ and $j = 1, \dots, p$), we select the numbers of principal components $u^{(r)}$ and $u^{j(r)}$ by minimizing the following Bayesian-type information criterion ([Cho and Qu \(2015\)](#)):

$$(13) \quad \Psi(u) = \frac{\text{tr}\{\mathbf{\Omega} - \tilde{\mathbf{\Omega}}(u)\}}{\text{tr}\{\mathbf{\Omega}\}} + u \frac{\log(n_r d)}{n_r d},$$

where $\text{tr}\{\cdot\}$ is the trace of a matrix, $\mathbf{\Omega}$ represents the weighting matrix $\mathbf{W}^{(r)}$ or $\mathbf{W}^{j(r)}$ with dimension d , $\tilde{\mathbf{\Omega}}(u) = \sum_{k=1}^u \xi_k \mathbf{v}_k \mathbf{v}_k^T$ is an approximation of $\mathbf{\Omega}$ based on the largest u eigenvectors, and ξ_k is the k th largest eigenvalue of $\mathbf{\Omega}$ corresponding to the eigenvector $\mathbf{v}_k$. Since $\text{tr}\{\mathbf{\Omega} - \tilde{\mathbf{\Omega}}(u)\} = \sum_{k=u+1}^d \xi_k$, the minimizer of $\Psi(u)$ is indeed the number of eigenvalues greater than $\text{tr}\{\mathbf{\Omega}\} \log(n_r d) / (n_r d)$.

Since the nonsmooth and nonseparable of fusion penalty $P_2(\Theta)$ in (10) makes the optimization challenging, we use the smoothing technique from [Chen et al. \(2012\)](#). Specifically, we reformulate the fused lasso as $P_2(\Theta) = \lambda_2 \|\mathbb{D}\Theta\|_1 = \lambda_2 \max_{\|\eta\|_\infty \leq 1} (\eta^T \mathbb{D}\Theta)$, where

$\mathbb{D}$ is a difference operator corresponding to the differences in $P_2(\Theta)$. Let $\tilde{P}_2(\Theta; \rho) = \lambda_2 \max_{\|\eta\|_\infty \leq 1} (\eta^T \mathbb{D}\Theta - \rho \|\eta\|_2^2 / 2)$, where ρ is a positive smoothing parameter and $\tilde{P}_2(\Theta; \rho)$ approximates $P_2(\Theta)$ as $\rho \rightarrow 0$. Define a projection operator $S(x) = -\mathbb{I}(x < -1) + x\mathbb{I}(-1 \leq$

Algorithm 1

-
- 1: LMI. Conduct longitudinal multiple imputation.
 - 2: Initialization. Obtain initial values $\Theta^{(0)}$ based on the averaged imputed data. Set the regularization parameters λ_1, λ_2 , and λ_3 .
 - 3: At the s th iteration, given $\Theta^{(s-1)}$ from the $(s-1)$ th iteration:
 - (a) PCA. For $r = 1, \dots, R$ and $j = 1, \dots, p$, select the number of principal components using (13) if $\mathbf{W}^{(r)}(\theta^{(s-1)})$ or $\mathbf{W}^{j(r)}(\vartheta^{j(s-1)})$ is singular or close to singular, and obtain $\mathbf{U}^{(r)}(\theta^{(s-1)})$, $\mathbf{U}^{j(r)}(\vartheta^{j(s-1)})$, $\tilde{\mathbf{W}}^{(r)}(\theta^{(s-1)})$, and $\tilde{\mathbf{W}}^{j(r)}(\vartheta^{j(s-1)})$.
 - (b) Optimization. Solve (15) through the gradient descent to obtain $\Theta^{(s)}$.
 - 4: Iterate Step 3 until $\|\Theta^{(s)} - \Theta^{(s-1)}\|_1 < 10^{-6}$. Return $\Theta^{(s)}$.
-

$x \leq 1) + \mathbb{I}(x > 1)$; we can deduce that $\eta^* = S(\mathbb{D}\Theta/\rho)$ is indeed the optimal solution in $\tilde{P}_2(\Theta; \rho)$. Thus, we have

$$(14) \quad \tilde{P}_2(\Theta, \rho) = \lambda_2 \left(\eta^{*T} \mathbb{D}\Theta - \frac{\rho}{2} \|\eta^*\|_2^2 \right),$$

which is convex and differentiable in Θ . We set $\rho = 10^{-4}$ following Chen et al. (2012).

We propose an algorithm based on the iterated GMM with PCA. Specifically, given $\Theta^{(s-1)}$ from the $(s-1)$ th iteration, for $r = 1, \dots, R$ and $j = 1, \dots, p$, if the weighting matrix $\mathbf{W}^{(r)}(\theta^{(s-1)})$ or $\mathbf{W}^{j(r)}(\vartheta^{j(s-1)})$ is singular or close to singular, we first select the number of principal components $u^{(r)(s-1)}$ and $u^{j(r)(s-1)}$ by minimizing the information criterion in (13), then construct principal component matrices $\mathbf{U}^{(r)}(\theta^{(s-1)})$ and $\mathbf{U}^{j(r)}(\vartheta^{j(s-1)})$, next get the reduced moment conditions $\tilde{\mathbf{g}}^{(r)}(\theta^{(s-1)})$ and $\tilde{\mathbf{h}}^{j(r)}(\vartheta^{j(s-1)})$, and finally, obtain the transformed weighting matrices $\tilde{\mathbf{W}}^{(r)}(\theta^{(s-1)})$ and $\tilde{\mathbf{W}}^{j(r)}(\vartheta^{j(s-1)})$.

At the s th iteration, we solve

$$(15) \quad \Theta^{(s)} = \underset{\Theta}{\operatorname{argmin}} \tilde{L}^{(s)}(\Theta),$$

where $\tilde{L}^{(s)}(\Theta) = \tilde{G}^{(s-1)}(\theta) + \sum_{j=1}^p \tilde{H}^{j(s-1)}(\vartheta^j) + P_1(\Theta) + \tilde{P}_2(\Theta; \rho) + P_3(\Theta)$. The quadratic form $\tilde{G}^{(s-1)}(\theta) = \sum_{r=1}^R (\tilde{\mathbf{g}}^{(r)(s-1)}(\theta))^T (\tilde{\mathbf{W}}^{(r)}(\theta^{(s-1)}))^{-1} \tilde{\mathbf{g}}^{(r)(s-1)}(\theta)$ with $\tilde{\mathbf{g}}^{(r)(s-1)}(\theta) = \mathbf{U}^{(r)}(\theta^{(s-1)}) \mathbf{g}^{(r)}(\theta)$ uses principal component matrix $\mathbf{U}^{(r)}(\theta^{(s-1)})$ and transformed weighting matrix $\tilde{\mathbf{W}}^{(r)}(\theta^{(s-1)})$ obtained at the $(s-1)$ th iteration. The quadratic form $\tilde{H}^{j(s-1)}(\vartheta^j) = \sum_{r=1}^R (\tilde{\mathbf{h}}^{j(r)(s-1)}(\vartheta^j))^T (\tilde{\mathbf{W}}^{j(r)}(\vartheta^{j(s-1)}))^{-1} \tilde{\mathbf{h}}^{j(r)(s-1)}(\vartheta^j)$ uses principal component matrix $\mathbf{U}^{j(r)}(\vartheta^{j(s-1)})$ and transformed weighting matrix $\tilde{\mathbf{W}}^{j(r)}(\vartheta^{j(s-1)})$ obtained at the $(s-1)$ th iteration. The above calculation process can be summarized in Algorithm 1, and an expanded version of this algorithm is shown in the Supplementary Material.

The turning parameter a in (9) is set to be 3.7 following Fan and Li (2001), and b in (9) is set to be 10 following Xue et al. (2022). To determine the regularization parameters $\lambda = (\lambda_1, \lambda_2, \lambda_3)^T$, we propose the following Bayesian-type information criterion (Xue and Qu (2021), Xue et al. (2022)):

$$(16) \quad \Phi(\lambda_1, \lambda_2, \lambda_3) = \log \left\{ \frac{\text{RSS}_Y(\hat{\Theta}_\lambda)}{NT} \right\} + \log \left\{ \sum_{j=1}^p \frac{\text{RSS}_{M^j}(\hat{\Theta}_\lambda)}{NT} \right\} + \frac{\log(NT)}{NT} \text{df}_\lambda,$$

where $\text{RSS}_Y(\hat{\Theta}_\lambda) = \sum_{r=1}^R \sum_{i \in \mathcal{H}(r)} \sum_{t=1}^T \frac{1}{|\mathcal{T}(r,t)|} \sum_{t \in \mathcal{T}(r,t)} (Y_{it} - \hat{\alpha}_t X_{it} - \hat{\beta}_t^T \mathbf{M}_{it}^{(t)} - \hat{\delta}^T \mathbf{Z}_i)^2$ and $\text{RSS}_{M^j}(\hat{\Theta}_\lambda) = \sum_{r=1}^R \sum_{i \in \mathcal{H}(r)} \sum_{t=1}^T \frac{1}{|\mathcal{T}(r,t)|} \sum_{t \in \mathcal{T}(r,t)} (M_{it}^{j(t)} - \hat{\gamma}_t^j X_{it} - \hat{\mathbf{f}}_t^{jT} \mathbf{Z}_i)^2$ are the residual sum of squares based on imputed data corresponding to models (1) and (2).

Degrees of freedom $\text{df}_\lambda = \text{df}_\lambda(\hat{\alpha}) + \text{df}_\lambda(\hat{\beta}) + \text{df}_\lambda(\hat{\gamma}) + \text{df}_\lambda(\hat{\delta}, \hat{\Gamma})$, where $\text{df}_\lambda(\hat{\delta}, \hat{\Gamma})$ is the number of nonzero estimated values in $\hat{\delta}$ and $\hat{\Gamma}$, and

$$\begin{aligned}\text{df}_\lambda(\hat{\alpha}) &= T - \sum_{t=1}^T \mathbb{I}(\hat{\alpha}_t = 0) - \sum_{t=2}^T \mathbb{I}(\hat{\alpha}_t = \hat{\alpha}_{t-1} \text{ and } \hat{\alpha}_t \hat{\alpha}_{t-1} \neq 0), \\ \text{df}_\lambda(\hat{\beta}) &= \sum_{j=1}^p \left\{ T - \sum_{t=1}^T \mathbb{I}(\hat{\beta}_t^j = 0) - \sum_{t=2}^T \mathbb{I}(\hat{\beta}_t^j = \hat{\beta}_{t-1}^j \text{ and } \hat{\beta}_t^j \hat{\beta}_{t-1}^j \neq 0) \right\}, \\ \text{df}_\lambda(\hat{\gamma}) &= \sum_{j=1}^p \left\{ T - \sum_{t=1}^T \mathbb{I}(\hat{\gamma}_t^j = 0) - \sum_{t=2}^T \mathbb{I}(\hat{\gamma}_t^j = \hat{\gamma}_{t-1}^j \text{ and } \hat{\gamma}_t^j \hat{\gamma}_{t-1}^j \neq 0) \right\}\end{aligned}$$

are numbers of nonzero and time-varying estimated values in $\hat{\alpha} = (\hat{\alpha}_1, \dots, \hat{\alpha}_T)^T$, $\hat{\beta} = (\hat{\beta}_1^T, \dots, \hat{\beta}_T^T)^T$, and $\hat{\gamma} = (\hat{\gamma}_1^T, \dots, \hat{\gamma}_T^T)^T$, respectively.

REMARK 4. There are three regularization parameters $(\lambda_1, \lambda_2, \lambda_3)$ that need to be selected during computation, and three-dimensional searches may be computationally costly. In practical applications we can adjust for confounders $\mathbf{Z}_i$ in models (1) and (2) during the data preprocessing stage (Schuermans, Dunn and Lussier (2024)) and then use the residuals for subsequent analyses. This approach removes the need to include parameters δ and Γ in the model, thereby eliminating the necessity of selecting λ_3 . For the remaining parameters (λ_1, λ_2) , we perform a two-dimensional grid search to identify the combinations which minimize the criterion (16).

6. Simulations. In this section we conduct comprehensive simulation studies to compare the performance of proposed method with some existing approaches. We consider the following time-varying structural equation models:

$$(17) \quad Y_{it} = \alpha_t X_{it} + \beta_t^T \mathbf{M}_{it} + e_{it},$$

$$(18) \quad \mathbf{M}_{it} = \gamma_t X_{it} + \eta_t Y_{i,t-1} + \epsilon_{it},$$

with $q = 0$ and $T = 4$. We generate the exposure $X_{i1} \sim \text{Uniform}(0, 1)$ and $X_{it} = X_{i,t-1} + \text{Uniform}(0, 1)$ for $t = 2, \dots, T$. The random error vector $(e_{i1}, \dots, e_{iT})^T$ is generated from the multivariate normal distribution $\mathcal{N}(\mathbf{0}, \Sigma(\rho_1))$, where $\Sigma(\rho_1)$ is the exchangeable correlation structure with parameter ρ_1 . The random error vector $(\epsilon_{i1}^T, \dots, \epsilon_{iT}^T)^T = (\epsilon_{i1}^1, \dots, \epsilon_{i1}^p, \dots, \epsilon_{iT}^1, \dots, \epsilon_{iT}^p)^T \sim \mathcal{N}(\mathbf{0}, \Sigma(\rho_2))$, where $\Sigma(\rho_2)$ is the blockwise compound symmetry correlation structure with parameter ρ_2 . We set $\rho_1 = 0.1$, $\rho_2 = 0.5$ and $\eta_t = \mathbf{0}$.

We consider the following settings with different sample sizes, different dimensions of mediators, different nonmonotone missingness structures, and different missingness mechanisms:

Setting I. We consider the missingness structure (A) illustrated in Figure S1 of the Supplementary Material. We set $N = 350$, $p = 50$, and the sample sizes in groups 1–4 are (50, 100, 100, 100). The true parameters $\alpha = (0.5, 0.5, 1, 1)^T$, β_t^j and γ_t^j for $j = 1, \dots, 14$ and $t = 1, \dots, 4$ are shown in Figure S2 of the Supplementary Material. The β_t^j and γ_t^j are both zero for $j = 15, \dots, 50$ and all t .

Setting II. We consider the missingness structure (B) illustrated in Figure S1 of the Supplementary Material, where there is no complete case group. We set $N = 375$, $p = 50$, and the sample sizes in groups 1–6 are (75, 75, 75, 50, 50, 50). The true parameters $\alpha = (0.5, 0.5, 1, 1)^T$, β_t^j and γ_t^j for $j = 1, \dots, 14$ and $t = 1, \dots, 4$ are shown in Figure S2 of the Supplementary Material. The β_t^j and γ_t^j are both zero for $j = 15, \dots, 50$ and all t .

Setting III. We consider the missingness structure (A) illustrated in Figure S1 of the Supplementary Material. We set $N = 550$, $p = 200$, and the sample sizes in groups 1–4 are $(100, 150, 150, 150)$. The true parameters $\alpha = (1, 1, 1, 1)^T$, β_t^j and γ_t^j for $j = 1, \dots, 22$ and $t = 1, \dots, 4$ are shown in Figure S3 of the Supplementary Material. The β_t^j and γ_t^j are both zero for $j = 23, \dots, 200$ and all t .

Setting IV. We consider the missingness structure (B) illustrated in Figure S1 of the Supplementary Material, where there is no complete case group. We set $N = 675$, $p = 200$, and the sample sizes in groups 1–6 are $(125, 125, 125, 100, 100, 100)$. The true parameters $\alpha = (1, 1, 1, 1)^T$, β_t^j and γ_t^j for $j = 1, \dots, 22$ and $t = 1, \dots, 4$ are shown in Figure S3 of the Supplementary Material. The β_t^j and γ_t^j are both zero for $j = 23, \dots, 200$ and all t .

In each of the settings I–IV, we consider three different missingness mechanisms (Little and Rubin (2019)): missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR). In the MCAR scenario, all subjects are completely randomly assigned to the groups. In the MAR scenario, subjects $i = 1, \dots, N$ are sequentially randomly assigned to group 1 with probability proportional to $\sum_{t=1}^T |X_{it}|$. The remaining subjects $i \notin \mathcal{H}(1)$ are sequentially randomly assigned to group 2 with probability proportional to $\sum_{t=1}^T |X_{it}|$, and so on until all subjects are grouped. In the MNAR scenario, the grouping process is the same as that in the MAR scenario, except that the probability is proportional to $\sum_{j=1}^p \sum_{t=1}^T |M_{it}^j|$.

Our method, using varying-coefficient (VC) models (17) and (18) with LMI-data, is compared with other mediator selection approaches, including: (i) Mix: mixed-effects models with time interactions and lasso penalty (Rijnhart et al. (2022)); (ii) Path: linear structural equation models with pathway lasso penalty (Zhao and Luo (2022)); (iii) HIMA: linear structural equation models with de-biased lasso and false discovery rate control (Perera et al. (2022)); (iv) Bayes: Bayesian sparse models with continuous shrinkage (Song et al. (2020)). Note that, for VC and Mix, the parameters are estimated using data from all time points simultaneously, while for Path, HIMA, and Bayes, the parameters are estimated using data from each time point separately. We also consider different approaches for handling missing data, including: (i) CC: complete-case analysis which retains only subjects in group 1 with fully observed repeated measurements; (ii) MC: matrix completion via iterative soft-thresholded singular value decomposition (Mazumder, Hastie and Tibshirani (2010)); (iii) SI: single imputation based on trajectory means (Jahangiri et al. (2023)).

In each replication, to evaluate the accuracy of the mediator selection for each method, we calculate the false negative rate (FNR) $\frac{1}{T} \sum_{t=1}^T \frac{\sum_{j=1}^p \mathbb{I}(\beta_t^j \gamma_t^j \neq 0, \hat{\beta}_t^j \hat{\gamma}_t^j = 0)}{\sum_{j=1}^p \mathbb{I}(\beta_t^j \gamma_t^j \neq 0)}$ and the false positive rate (FPR) $\frac{1}{T} \sum_{t=1}^T \frac{\sum_{j=1}^p \mathbb{I}(\beta_t^j \gamma_t^j = 0, \hat{\beta}_t^j \hat{\gamma}_t^j \neq 0)}{\sum_{j=1}^p \mathbb{I}(\beta_t^j \gamma_t^j = 0)}$, across all time points, where $\hat{\beta}_t^j$ and $\hat{\gamma}_t^j$ are estimated values of β_t^j and γ_t^j for $t = 1, \dots, T$ and $j = 1, \dots, p$. We also calculate the mean squared error (MSE) $\sum_{t=1}^T \sum_{j=1}^p (\beta_t^j \gamma_t^j - \hat{\beta}_t^j \hat{\gamma}_t^j)^2 / T$ to evaluate the precision of the mediation effect estimation for each method. We say that a method has better performance if it has a smaller FNR+FPR and a smaller MSE.

Table 1 summarizes the averaged estimation results across 50 replications of different methods for different missingness mechanisms under setting I. Table 1 shows that the proposed VC-LMI method has the smallest FNR+FPR and MSE among all missingness mechanisms, which suggests that the proposed VC-LMI method has the highest accuracy of mediator selection and precision of mediation effect estimation, and its performance is robust to different missingness mechanisms. The VC-CC has a relatively large FNR+FPR, due to the fact that only 50/350 of the subjects with complete repeated measurements are used for analysis. The VC-MC has large FNR+FPR and MSE, possibly because the MC assumes the

TABLE 1
False negative rate (FNR), false positive rate (FPR), FNR+FPR, and mean squared error (MSE) of different methods for different missingness mechanisms under setting I

	Missing completely at random				Missing at random				Missing not at random			
	FNR	FPR	FNR+FPR	MSE	FNR	FPR	FNR+FPR	MSE	FNR	FPR	FNR+FPR	MSE
VC-LMI	0.059	0.086	0.145	0.719	0.049	0.090	0.139	0.774	0.064	0.088	0.152	0.715
VC-CC	0.174	0.054	0.228	1.032	0.180	0.057	0.237	1.018	0.150	0.058	0.208	0.860
VC-MC	0.419	0.072	0.491	8.332	0.333	0.074	0.407	8.590	0.401	0.071	0.471	8.176
VC-SI	0.203	0.036	0.239	5.065	0.180	0.034	0.214	5.228	0.196	0.036	0.232	5.208
Mix-CC	0.111	0.105	0.215	2.638	0.109	0.114	0.222	2.525	0.116	0.099	0.215	2.319
Mix-MC	0.086	0.327	0.414	8.815	0.077	0.320	0.397	9.214	0.092	0.314	0.407	8.772
Mix-SI	0.057	0.167	0.224	5.839	0.043	0.163	0.205	6.016	0.049	0.157	0.206	5.965
Path-CC	0.272	0.065	0.337	7.215	0.288	0.064	0.351	7.125	0.258	0.065	0.323	7.082
Path-MC	0.560	0.032	0.592	10.420	0.470	0.028	0.498	10.364	0.541	0.033	0.574	10.382
Path-SI	0.248	0.020	0.268	9.198	0.227	0.022	0.250	9.319	0.249	0.021	0.270	9.285
HIMA-CC	0.679	0.005	0.685	1.715	0.702	0.004	0.707	2.198	0.669	0.004	0.673	1.708
HIMA-MC	0.574	0.035	0.609	8.742	0.489	0.035	0.524	9.173	0.547	0.040	0.587	8.579
HIMA-SI	0.376	0.014	0.390	4.895	0.361	0.015	0.377	5.066	0.370	0.015	0.385	5.158
Bayes-CC	0.916	0.000	0.916	5.577	0.916	0.000	0.916	5.508	0.901	0.000	0.901	4.797
Bayes-MC	0.786	0.019	0.805	10.043	0.754	0.021	0.775	10.840	0.760	0.020	0.780	9.711
Bayes-SI	0.689	0.001	0.690	6.129	0.675	0.001	0.676	6.129	0.659	0.001	0.660	6.115

Note: Each estimator is denoted as “mediator selection method—missing data handling method.” Results are averaged across all time points and 50 replications

low-rank structure and noisy entries of the variable matrix, which may not be suitable for our longitudinal data. The VC-SI also has large FNR+FPR and MSE, possibly because the SI utilizes the trajectory mean for imputation, which erases some of the longitudinal changes in variables. Our VC-LMI method makes full use of the correlation among repeated measurements for imputation and effectively integrates the multiple imputations, thereby exhibiting powerful performance compared with candidate approaches.

For other mediator selection methods in Table 1, both FNR+FPR and MSE are large. Specifically, the Mix shows large FPR, while the Path provides large FNR. The reason for their poor estimations may be because the mediator selection methods with some shortcomings (too many interactions in Mix and overshrinkage in Path) are used on the datasets with poor quality. The HIMA and Bayes provide a smaller FPR but a larger FNR, possibly due to the multiple-testing procedure for HIMA and the posterior inclusion probability for Bayes being overly conservative.

Table 2 summarizes the estimation results under setting II. Since no subject has complete repeated measurements, the CC is not applicable. Our proposed VC-LMI method has the smallest FNR+FPR and MSE among all candidate approaches. To evaluate the performance of the proposed method with a greater number of time points, we conduct simulations with $T = 4$ and $T = 5$, with $p = 50, 40$ subjects in each group and the missingness structure illustrated in Figure S4 of the Supplementary Material. We also compare the proposed method with the cross-sectional mediation method of Xue et al. (2022) (CS), which analyzes data collected at each time point separately. Table 3 summarizes the results under the MAR mechanism, demonstrating that our VC-LMI method outperforms other approaches in terms of mediator selection accuracy and mediation effect estimation precision. However, the computation time for one replication is 1582 seconds for $T = 4$ and 6504 seconds for $T = 5$, which is longer than that of other approaches.

TABLE 2

False negative rate (FNR), false positive rate (FPR), FNR+FPR, and mean squared error (MSE) of different methods for different missingness mechanisms under setting II

	Missing completely at random				Missing at random				Missing not at random			
	FNR	FPR	FNR+FPR	MSE	FNR	FPR	FNR+FPR	MSE	FNR	FPR	FNR+FPR	MSE
VC-LMI	0.046	0.105	0.151	0.790	0.048	0.113	0.160	0.855	0.056	0.107	0.163	0.755
VC-MC	0.439	0.076	0.514	9.195	0.369	0.070	0.439	9.082	0.426	0.072	0.498	8.962
VC-SI	0.254	0.043	0.297	5.426	0.276	0.043	0.319	5.477	0.246	0.043	0.290	5.374
Mix-MC	0.105	0.330	0.435	10.002	0.091	0.291	0.382	10.131	0.114	0.297	0.411	9.880
Mix-SI	0.057	0.140	0.198	6.707	0.060	0.143	0.203	6.606	0.056	0.151	0.207	6.581
Path-MC	0.646	0.022	0.668	10.908	0.559	0.017	0.577	10.770	0.611	0.018	0.630	10.851
Path-SI	0.286	0.019	0.305	9.609	0.276	0.016	0.292	9.534	0.264	0.019	0.283	9.544
HIMA-MC	0.654	0.045	0.698	9.667	0.571	0.046	0.618	9.837	0.631	0.048	0.680	9.564
HIMA-SI	0.457	0.021	0.478	5.321	0.461	0.018	0.479	5.287	0.429	0.021	0.450	5.360
Bayes-MC	0.915	0.028	0.943	10.791	0.876	0.031	0.907	11.405	0.902	0.030	0.933	10.800
Bayes-SI	0.756	0.008	0.764	5.960	0.774	0.005	0.780	6.075	0.764	0.006	0.770	6.010

Note: Each estimator is denoted as “mediator selection method—missing data handling method.” Results are averaged across all time points and 50 replications

TABLE 3

False negative rate (FNR), false positive rate (FPR), FNR+FPR, and mean squared error (MSE) of different methods for missing at random mechanism, with $T = 4$ or $T = 5$ time points

	$T = 4$ time points					$T = 5$ time points				
	FNR	FPR	FNR+FPR	MSE	Time	FNR	FPR	FNR+FPR	MSE	Time
VC-LMI	0.019	0.155	0.174	0.703	1582.174	0.015	0.138	0.154	0.581	6503.613
VC-CC	0.224	0.063	0.287	1.269	54.628	0.212	0.049	0.261	0.875	101.699
VC-MC	0.298	0.074	0.371	8.919	39.083	0.498	0.060	0.558	4.704	242.354
VC-SI	0.226	0.032	0.258	5.445	25.953	0.233	0.030	0.263	4.001	87.640
CS-CC	0.214	0.080	0.294	1.487	6.244	0.226	0.056	0.282	1.014	7.661
CS-MC	0.312	0.075	0.387	8.909	33.188	0.446	0.066	0.512	4.633	145.474
CS-SI	0.190	0.040	0.230	5.311	19.402	0.165	0.033	0.198	3.835	42.083
Mix-CC	0.136	0.098	0.234	3.102	3.517	0.077	0.104	0.181	2.508	3.740
Mix-MC	0.060	0.342	0.402	9.707	6.064	0.043	0.328	0.370	4.847	12.710
Mix-SI	0.019	0.161	0.180	6.143	5.367	0.001	0.184	0.185	3.811	8.980
Path-CC	0.305	0.085	0.390	7.468	20.683	0.237	0.083	0.321	5.963	24.242
Path-MC	0.543	0.017	0.560	10.690	17.222	0.388	0.002	0.390	8.407	24.842
Path-SI	0.207	0.012	0.219	9.475	16.518	0.150	0.013	0.163	7.586	20.044
HIMA-CC	0.721	0.006	0.728	3.465	18.991	0.736	0.005	0.741	1.794	23.598
HIMA-MC	0.381	0.051	0.432	9.343	19.529	0.387	0.052	0.439	4.799	29.222
HIMA-SI	0.279	0.024	0.303	5.129	17.851	0.122	0.033	0.155	3.438	24.593
Bayes-CC	0.936	0.000	0.936	6.371	11.440	0.947	0.000	0.947	5.484	14.111
Bayes-MC	0.779	0.034	0.813	10.359	48.438	0.683	0.035	0.719	4.844	114.912
Bayes-SI	0.564	0.012	0.576	5.478	47.479	0.319	0.018	0.337	3.527	110.883

Note: Each estimator is denoted as “mediator selection method—missing data handling method.” “Time” represents the computational time (in seconds) of each method for one replication with given tuning parameters. Results are averaged across all time points and 50 replications

The estimation results under settings III and IV in the high-dimensional context are provided in Tables S1 and S2 of the Supplementary Material, which show that our proposed VC-LMI method also beats other candidate approaches, based on its smallest FNR+FPR and MSE. The Supplementary Material includes additional simulations, where Table S3 demonstrates that our method continues to outperform other candidate approaches in higher-dimensional settings ($p = 2000$), Table S4 shows that combining multiple imputations through the GMM outperforms approach that selecting imputations with the smallest variance.

7. DNHS data analysis. In this section we investigate how DNAm dynamically mediates the effect of traumatic experiences on development of PTSD, utilizing the DNHS data. Specifically, we treat the number of traumatic events as exposure X_{it} , the logarithm transformed PTS symptom severity score as outcome Y_{it} , and the M-values (Kruppa et al. (2021)) of DNAm CpG sites as potential mediators M_{it} . The study includes 526 subjects in waves 1, 2, 4, and 5, which are referred to as time points 1–4. In our data, exposure X_{it} and outcome Y_{it} are always observed, but the mediators M_{it} suffer from nonmonotone missingness, as shown in Figure 1: M_i of some subjects may be available at certain visits, but missing at the next time point, and measured again at later visits.

There are two steps in the data preprocessing phase. First, we adjust some demographic information and blood work measured at baseline. For the outcome corresponding to model (1), we adjust age, gender, race, education, income, cigarette use, depression, and social cohesion score (Johns et al. (2012), Wani et al. (2021)). For the mediators corresponding to model (2), we adjust age, gender, cigarette use, CD4 T cells, CD8 T cells, natural killer cells, B cells, and monocytes (Occean et al. (2022)). The residuals after adjustment and the models (17) and (18) are used for subsequent analyses. Second, given the limited number of subjects and the ultra-high-dimensional potential mediators (1879 DNAm CpG sites), we carry out a screening process to reduce the dimension to a moderate scale below the sample size (Fan and Lv (2008)). Specifically, for $t = 1, \dots, T$ and $j = 1, \dots, p$, we consider a series of marginal models $Y_{it} = \alpha_t X_{it} + \beta_t^j M_{it}^j + e_{it}$ and $M_{it}^j = \gamma_t^j X_{it} + \eta_t^j Y_{i,t-1} + \varepsilon_{it}^j$. Along the lines of the sure independence mediator screening (Perera et al. (2022)), for each time point $t = 1, \dots, T$, we use the subjects with observed mediators to fit the marginal models, then obtain a subset $\mathcal{D}_t = \{j: M^j \text{ is among the top } [n_t/\log(n_t)] \text{ largest } \beta_t^j \gamma_t^j \text{ effect}\}$, where n_t is the sample size at time point t and $[\cdot]$ denotes the integer part. Then we take $\bigcup_{t=1, \dots, T} \mathcal{D}_t$ as the index set of potential mediators after screening. There are 153 DNAm CpG sites remaining for the next step of analysis.

We use the proposed VC-LMI method to fit the preprocessed data. The estimated direct effect is $(\hat{\alpha}_1, \hat{\alpha}_2, \hat{\alpha}_3, \hat{\alpha}_4) = (0.50, 0.54, 0.54, 0.54)$, which means that trauma exposure has a sustained direct effect on PTSD. Figures 4 and 5 show the 13 selected CpG sites, each of which has nonzero $\hat{\beta}_t^j \hat{\gamma}_t^j$ at least at one time point. We can find that certain CpG sites (e.g., cg22564046) initially have nonzero mediation effects, but the effects disappear over time. Certain CpG sites (e.g., cg17318247) do not show mediation effects in the early stages, but the effects emerge over time. This indicates that the effects at the CpG level may vary in different stages of the disease process.

As shown in Figure 5, the estimated mediation effects for certain CpG sites at specific time points, such as cg22508145 at $t = 1, 2$ and cg09247979 at $t = 3, 4$, appear small or close to zero. These seemingly negligible effects should be interpreted within the broader context of longitudinal mediation analysis and the cumulative role of DNAm across multiple time points. First, mediation effects can vary over time due to the dynamic nature of DNAm and its interactions with environmental exposures and disease progression. Small mediation effects at individual time points do not necessarily indicate that the CpG site lacks relevance

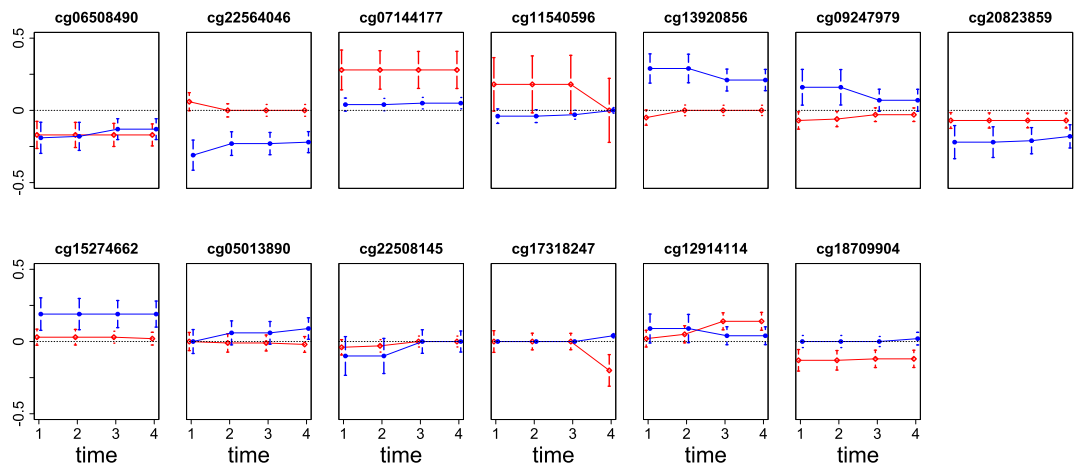


FIG. 4. Estimated values $\hat{\beta}_t^j$ (red colored) and $\hat{\gamma}_t^j$ (blue colored) with Bootstrap 95% confidence intervals for the selected CpG sites and $t = 1, \dots, 4$, based on the proposed VC-LMI method.

as a mediator. Instead, they may reflect temporal fluctuations or periods during which the indirect effect is less pronounced. Second, while mediation effects may appear negligible at specific time points, the cumulative contribution of a CpG site across all time points could still be meaningful. Longitudinal analyses are designed to capture these aggregate patterns, which may not be evident when focusing solely on isolated single-time-point effects. Finally, even small mediation effects can hold biological significance, especially in high-dimensional settings where individual mediators contribute subtly but collectively to disease mechanisms (Xue et al. (2022)).

Table 4 lists the gene information corresponding to the selected CpG sites. Eleven of these sites are annotated to human genes according to the Illumina annotation files (*CAT*, *PFKP*, *EEF1E1*, *TANC1*, *PTPRK*, *OVGP1*, *RPS6KA2*, *CPAMD8*, *CDK16*, *FAM120B*, and *C14orf182*). Of these 11 genes, *PTPRK* has been shown to be associated with mental health disorders such as PTSD (Chitrula, Nagarkatti and Nagarkatti (2016)); *RPS6KA2* has been reported as significant KEGG pathways for PTSD in epigenome-wide DNA methylation study (Kuan et al. (2017)). Other CpG sites, while not located near any protein-coding genes, fall

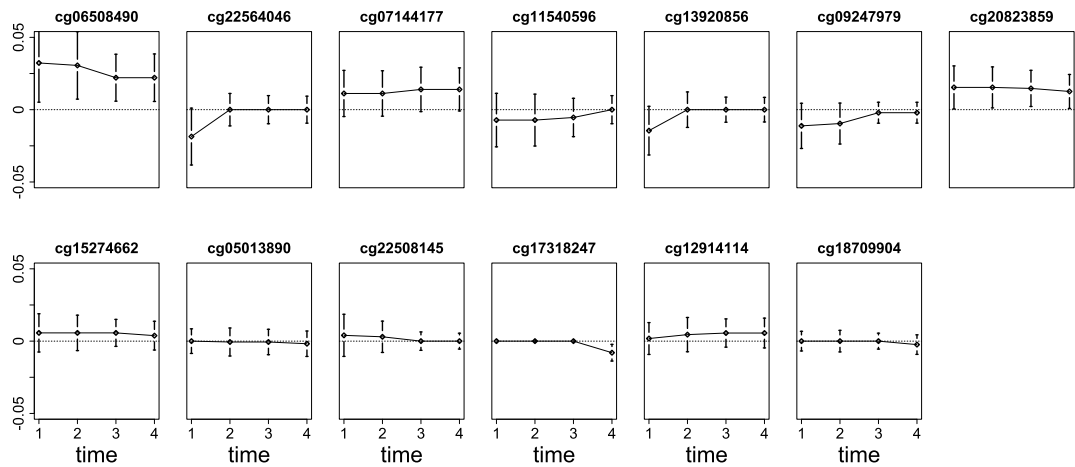


FIG. 5. Estimated mediation effects $\hat{\beta}_t^j \hat{\gamma}_t^j$ with Bootstrap 95% confidence intervals for the selected CpG sites and $t = 1, \dots, 4$, based on the proposed VC-LMI method.

TABLE 4
Gene information corresponding to the selected CpG sites

CpG	Chromosome	Gene
cg06508490	11	CAT
cg22564046	10	PFKP
cg07144177	6	EEF1E1
cg11540596	6	NA/Desert
cg13920856	2	TANC1
cg09247979	6	PTPRK
cg20823859	1	OVGP1
cg15274662	7	NA/Desert
cg05013890	6	RPS6KA2
cg22508145	19	CPAMD8
cg17318247	X	CDK16
cg12914114	6	FAM120B
cg18709904	14	C14orf182

within ENCODE candidate Cis-Regulatory Elements that are adjacent to highly conserved noncoding genomic regions (Moore et al. (2020)).

Beyond the time-invariant associations, our results further show that the effects of genes change over time, which may reveal how the biological embedding of trauma exposure through DNAm contributes to PTSD risk over development. Our findings could have important clinical implications for time-sensitive risk prediction in trauma-exposed populations.

To compare the performance of the proposed VC-LMI method with other approaches, we randomly split the preprocessed data into a training set (80% of subjects) and a testing set (20% of subjects; the index set is denoted as $\mathfrak{T}$) for 50 replications. In each replication and for each method, we train the model on the training set and obtain the estimated values of parameters. Then we calculate the mean number of selected mediators (No.-mediator) as $\frac{1}{T} \sum_{t=1}^T \sum_{j=1}^P \mathbb{I}(\hat{\beta}_t^j \hat{\gamma}_t^j \neq 0)$, the prediction mean squared error (PMSE- β) for the outcome model (17) as $\frac{1}{|\mathfrak{T}|} \sum_{i \in \mathfrak{T}} \sum_{r=1}^R \mathbb{I}(i \in \mathcal{H}(r)) \frac{1}{|o(r)|} \sum_{t \in o(r)} (Y_{it} - \hat{Y}_{it})^2$, and the prediction mean squared error (PMSE- γ) for the mediator model (18) as $\frac{1}{|\mathfrak{T}|p} \sum_{i \in \mathfrak{T}} \sum_{r=1}^R \mathbb{I}(i \in \mathcal{H}(r)) \frac{1}{|o(r)|} \times \sum_{t \in o(r)} (\mathbf{M}_{it} - \hat{\mathbf{M}}_{it})^T (\mathbf{M}_{it} - \hat{\mathbf{M}}_{it})$, on the testing set across all time points, where $\hat{Y}_{it}$ and $\hat{\mathbf{M}}_{it}$ are the fitted values of Y_{it} and $\mathbf{M}_{it}$ for $i \in \mathfrak{T}$ and $t \in o(r)$.

Table 5 summarizes the averaged estimation results across 50 replications. We observe that our VC-LMI method has the smallest PMSE both in the outcome model and the mediator model, indicating that the proposed method is more accurate in terms of prediction. For the CS, Mix, and Path, despite more mediators being selected based on the imputed dataset, the PMSE is larger, possibly due to overfitting. For the HIMA and the Bayesian approaches, no mediator passes their overly conservative significance tests, which is consistent with the simulation results in Section 6.

8. Discussion. In this paper, to investigate the time-varying mediation effects of DNAm on the relationship between trauma and PTSD, we propose time-varying structural equation models. The LMI approach is proposed to handle nonmonotone missing DNAm and the regularization approach is incorporated to select relevant mediators and capture time-varying effects. In simulations the proposed method compares favorably against existing competitors in various scenarios. In DNHS data analysis, we successfully identify potential DNAm CpG sites which show dynamic mediation effects.

TABLE 5

Mean number of selected mediators (No.-mediator), prediction mean squared error (PMSE-β) for the outcome model (17), and prediction mean squared error (PMSE-γ) for the mediator model (18) in DNHS data analysis

	No.-mediator	PMSE-β	PMSE-γ	Time
VC-LMI	12.100	0.076	1.250	8592.738
VC-CC	9.680	0.350	1.807	294.837
VC-MC	15.720	0.092	1.251	92.966
VC-SI	23.860	0.101	1.257	135.713
CS-CC	10.360	0.504	1.810	14.061
CS-MC	18.360	0.089	1.251	32.906
CS-SI	24.320	0.100	1.257	46.951
Mix-CC	3.450	0.133	1.423	9.588
Mix-MC	5.960	0.093	1.252	115.027
Mix-SI	28.390	0.095	1.253	34.431
Path-CC	11.880	0.126	1.422	209.357
Path-MC	43.210	0.076	1.251	107.226
Path-SI	47.890	0.090	1.254	74.129
HIMA-CC	0.010	0.129	1.253	44.926
HIMA-MC	0.040	0.104	1.253	704.612
HIMA-SI	0.000	0.105	1.253	176.237
Bayes-CC	0.000	0.101	1.256	26.964
Bayes-MC	0.000	0.106	1.252	129.391
Bayes-SI	0.000	0.106	1.253	127.870

Note: Each estimator is denoted as “mediator selection method—missing data handling method.” “Time” represents the computational time (in seconds) of each method for one replication with given tuning parameters. Results are averaged across all time points and 50 replications

We assume the mediators ($M_{it}^1, \dots, M_{it}^p$) are correlated, but we do not specify causal ordering. The main reason for this assumption is that, based on the existing literature, external adverse environmental exposures or disease conditions are likely to influence DNAm across multiple CpG sites (Turecki and Meaney (2016)), whereas the relationships between DNAm levels across CpG sites are primarily correlational (Mou, Zhang and Arshad (2022)). Such assumptions of parallel relationships among mediators are commonly made in high-dimensional DNAm-related mediation analysis (e.g., Song et al. (2020), Perera et al. (2022), Xue et al. (2022)). Furthermore, even if causal ordering existed among high-dimensional DNAm CpG sites, establishing the correct causal order is infeasible in complex biological systems, instead they are characterized by intricate interactions and feedback loops (Tai et al. (2022)). In our study we also lack sufficient genetic knowledge to reliably support causal ordering. A more refined understanding of the causal relationships among high-dimensional mediators may require the application of directed acyclic graph learning techniques (Shi and Li (2022)), which might be able to identify the complex causal structures among variables. However, such analyses extend beyond the scope of this study and could be explored in future research.

Unmeasured confounding is an inherent challenge in population-based studies. In our DNHS data analysis, we incorporate a comprehensive set of demographic information and blood work measured at baseline to control for confounding bias as much as possible. Nonetheless, some unmeasured confounders, such as genetic predisposition (Rutten et al. (2018)), chronic health conditions and inflammation (Katrini et al. (2022)), detailed substance use history (Tortella-Feliu et al. (2019)), and psychosocial factors (Sibrava et al. (2019)), were not comprehensively measured in the DNHS data. As a result, residual con-

founding may still be present, which could potentially limit our study. To address these remaining unmeasured confounders, methods such as instrumental variable approaches (Chen et al. (2023)), proxy causal learning frameworks (Dukes, Shpitser and Tchetgen Tchetgen (2023)), or de-confounding techniques (Yuan and Qu (2024)) could be employed. Exploring these methodologies could be a promising direction for future research.

In our imputation step, we estimate $\mathbb{E}(M_t^j | \mathbf{M}_t, \mathbf{D})$ using specific groups that include observed M_t^j and $\mathbf{M}_t$, then impute the unobserved M_t^j in other groups that contain observed $\mathbf{M}_t$ but missing M_t^j . The rationale for our imputation method relies on the premise that the conditional distributions of mediators in the observed data are the same as those in the missing data, which holds only under the assumptions of MCAR or MAR. For mediators that are MNAR, the direct and indirect effects corresponding to the coefficients in models (1) and (2) are not identifiable without additional assumptions. Potential strategies include modeling the missing indicator $f_{R_i}(\mathbf{r}_i | X_i, Y_i, \mathbf{Z}_i, \mathbf{M}_i)$ based on the completeness assumption (Zuo et al. (2025)), or introducing instrumental or shadow variables to impose specific structures on the missingness mechanism (Shan, Li and Ai (2024)). Extending our method to accommodate MNAR mediators using these strategies as well as further addressing missing outcomes (Qin et al. (2019)) are important directions for future research.

From Table 3 and Table 5, we observe that, although our VC-LMI method outperforms other approaches in terms of mediator selection accuracy and mediation effect estimation precision, the computation time is greatly longer. There are two main sources of computational burden. The first arises during the imputation step, where a large number of regression models are fitted to capture the correlations among repeated measurements and multiple mediators. The second occurs during the GMM estimation step, which involves PCA and Newton-type optimization. The computational complexity of both steps depends on the number of mediators p and the number of missing blocks R . To improve computational efficiency, we can start by reducing p and R . For instance, the screening method, proposed by Fan and Lv (2008), can be used to reduce p , while mean imputation, instead of LMI, can be applied to blocks with fewer missing values to decrease R . Beyond these basic strategies, several advanced approaches offer additional potential for computational efficiency gains. For handling missing data, Xie, Xue and Wang (2024) proposed employing generative models for missing data imputation, using variational inference to substantially reduce computational costs compared to traditional sampling-based methods. Yu et al. (2020) proposed a direct sparse regression procedure that leverages covariance from block-missing data, utilizing an extended lasso-type estimator to obtain a sparse estimate of the coefficients in the optimal linear prediction. This procedure avoids imputation, resulting in a relatively low computational burden. For GMM estimation with high-dimensional parameters, Hector and Song (2021) developed a divide-and-conquer procedure implemented within a fully distributed and parallelized computational scheme, greatly reducing computing time in both the distributed and integrated steps. Exploring these methodologies to enhance computational efficiency could be potential directions for our future research.

For the statistical inference of the mediation effect, there are two main challenges including the limiting distributions of the estimators for parameters in models (1) and (2) and the composite p -value of the mediation effect. Specifically, the penalty functions in (9) and (10) may present challenges in deriving the asymptotic distribution. Some existing studies used the bi-level penalty similar to that of $P_1(\Theta)$ in (9), and the corresponding theoretical results show that the estimator of nonzero coefficients in general converges to nonnormal distribution (Huang et al. (2009)). Our estimator includes an additional fusion penalty $P_2(\Theta)$ in (10) to serve the purpose of shrinking similar effects at adjacent time points, which may lead to more complex limiting distributions. On the other hand, to test whether M^j mediates the effect of X on Y , the null and alternative hypotheses can be formulated as

$H_0^j : \beta^j \gamma^j = 0$ vs. $H_1^j : \beta^j \gamma^j \neq 0$. The null hypothesis encompasses a product of parameters, which is composite, and consists of three cases as case 1: $\beta^j = 0$ and $\gamma^j \neq 0$, case 2: $\beta^j \neq 0$ and $\gamma^j = 0$, and case 3: $\beta^j = 0$ and $\gamma^j = 0$. Constructing the test statistic and deriving the reference distribution and composite p -value under the composite null hypothesis are still open problems in the field of high-dimensional mediation analysis, and relevant studies can be found in the review by Du et al. (2023). Considering the complex limiting distributions of $\hat{\beta}^j$ and $\hat{\gamma}^j$, it is more challenge to conduct statistical inference for the mediation effect $\hat{\beta}^j \hat{\gamma}^j$. We will develop theories and inference tools of the proposed estimator in our future research.

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SUPPLEMENTARY MATERIAL

Supplement to “Time-varying mediation analysis for incomplete data with application to DNA methylation study for PTSD” (DOI: [10.1214/25-AOAS2076SUPPA](https://doi.org/10.1214/25-AOAS2076SUPPA); .pdf). The Supplementary Material includes three sections. Section 1 contains the GMM estimator corresponding to each mediator M^j ($j = 1, \dots, p$) in model (2). Section 2 contains the expanded version of Algorithm 1. Sections 3 and 4 contain the additional simulation results.

Supplemental code (DOI: [10.1214/25-AOAS2076SUPPB](https://doi.org/10.1214/25-AOAS2076SUPPB); .zip). We have implemented the proposed longitudinal multiple imputation method as an R function named “LMI” and the generalized method of moments with principal component analysis as another R function named “GMM_PCA”. Simulation code, data, and code for application study, along with detailed instructions, are provided in the supplementary file and are also available at https://github.com/Wei-Kecheng/long_med_miss.

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